

LEGACYLENS

AI-Assisted Modernization of Legacy .NET Systems for U.S. Critical Infrastructure

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Overview

LegacyLens is an open-source AI-assisted modernization platform designed to analyze, understand, secure, and modernize legacy .NET software systems used across healthcare, financial services, insurance, and government environments.

The platform combines compiler-level semantic analysis using Microsoft's Roslyn framework with advanced large language models to generate modernization recommendations that are technically accurate, auditable, and compliant with modern cybersecurity and regulatory requirements.

Problem

A significant portion of U.S. critical infrastructure continues to depend on aging software systems that are expensive to maintain, difficult to secure, and increasingly incompatible with modern cybersecurity standards.

Federal reports indicate that many public-sector systems remain decades old, while private-sector organizations face rising costs associated with technical debt, security vulnerabilities, and modernization delays.

Healthcare organizations experience some of the highest data breach costs of any industry, while financial institutions continue to face growing cybercrime risks and compliance challenges.

Proposed Solution

LegacyLens provides an AI-assisted modernization framework capable of:

- Analyzing legacy .NET applications
- Identifying obsolete technologies and security risks
- Extracting undocumented business rules
- Recommending modernization pathways
- Supporting migration toward secure cloud-native architectures
- Generating compliance-aware modernization guidance

The platform is being developed as an open-source project under the MIT License to maximize accessibility and adoption.

National Relevance

The project directly supports U.S. priorities in:

- Cybersecurity
- Critical infrastructure resilience
- Healthcare technology modernization
- Financial system security
- Federal technology modernization
- Software supply-chain security

Expected Impact

LegacyLens aims to reduce modernization costs, accelerate migration timelines, improve software security, preserve institutional knowledge embedded in legacy systems, and expand access to modernization expertise across organizations that lack dedicated modernization teams.

Current Status

- Technical Design Document completed
- Public GitHub repository established
- Architecture and implementation roadmap defined
- Phase 1 prototype under development

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